

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

Duration : 1 Hour

QUESTIONS: 5

Total Marks:50

Annexure A

Design Task

Code of Conduct:

I HEMASRINITHA P (192572179) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Hemasrinitha P

Design Task

1. Hybrid AI Recommendation Engine Design

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. Smart Disaster Detection System

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. AI-Based Fraud Detection Framework

Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. Autonomous Supply Chain Optimization System

Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. Personalized Learning Analytics Platform

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration

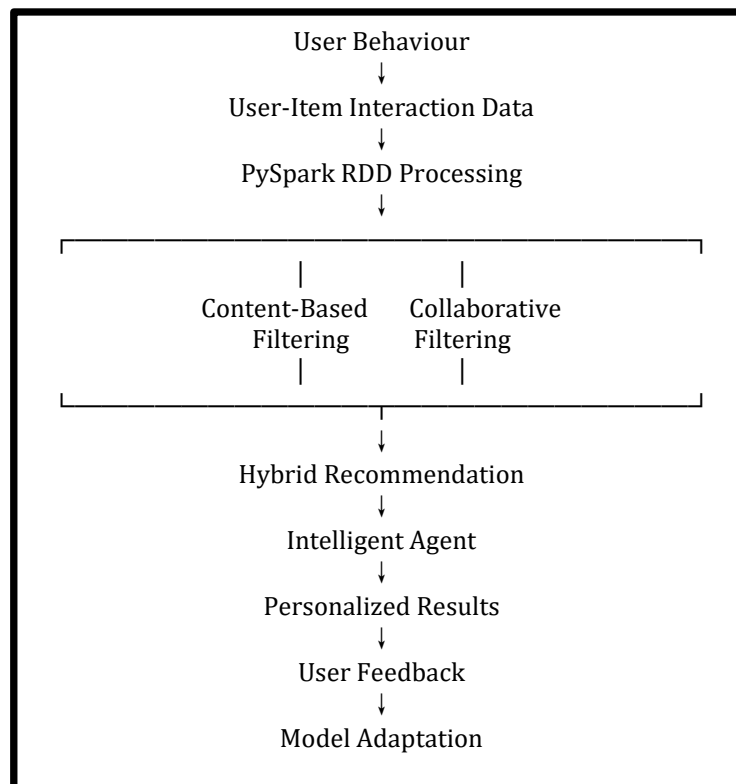
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation

QUESTION 1:-

HYBRID AI RECOMMENDATION ENGINE DESIGN

A hybrid recommendation engine combines Content-Based Filtering and Collaborative Filtering to provide more accurate and personalized recommendations. PySpark RDDs can be used to process large amounts of user-item interaction data in a distributed manner.

1. System Architecture



2. Content-Based Filtering

Content-based filtering recommends items that are similar to items the user has previously liked. For example, if a user frequently watches action movies, the system recommends other movies with similar genres, actors, or descriptions. PySpark RDDs can transform item features and calculate similarity between the user's preferences and available items.

3. Collaborative Filtering

Collaborative filtering uses the behaviour of multiple users. It identifies users with similar preferences and recommends items that similar users have liked. For example, if User A and User B have similar viewing patterns and User B likes a particular movie that User A has not watched, the system can recommend that movie to User A.

4. Use of PySpark RDDs

Large recommendation systems may contain millions of users and interactions. PySpark RDDs divide this data across multiple machines and process it in parallel. Operations such as `map()`, `filter()`, `groupByKey()` and `reduceByKey()` can be used to process user-item interactions efficiently. This reduces processing time and allows the system to scale to large datasets.

5. Hybrid Recommendation

The results from both filtering methods are combined to produce a final recommendation score.

Final Score = α (Content-Based Score) + β (Collaborative Score)

Here, α and β determine the importance given to each method. This approach reduces the limitations of using only one recommendation technique and can provide more accurate results.

6. Role of Intelligent Agents

An intelligent agent continuously observes user behaviour such as:

- Items viewed
- Items purchased
- Ratings given
- Items skipped
- Search history

The agent updates the user's preference profile based on this feedback. For example, if a user repeatedly ignores a particular type of recommendation, the agent can reduce its importance and provide more relevant alternatives.

7. Real-Time Adaptation

The system can continuously process new interactions and update recommendations. A user watching several science-fiction movies, for example, can immediately receive more science-fiction recommendations. Thus, the intelligent agent provides adaptive and personalized recommendations, while PySpark provides the distributed processing capability required for large-scale data.

Conclusion

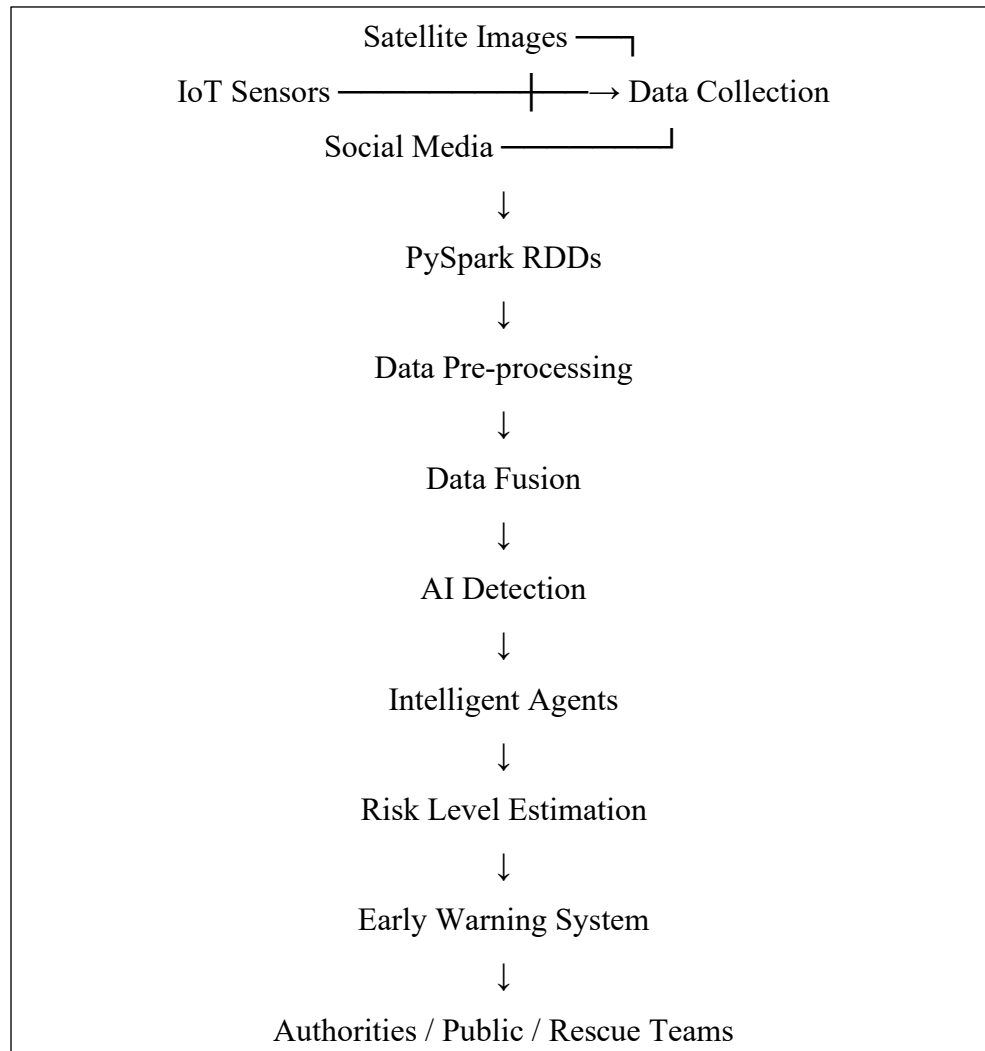
The proposed hybrid system combines the strengths of content-based and collaborative filtering. PySpark RDDs provide scalable distributed processing, while the intelligent agent continuously learns from user behaviour and feedback. This makes the recommendation engine suitable for large-scale applications such as e-commerce, movie platforms, music services, and online learning systems.

QUESTION 2:-

SMART DISASTER DETECTION SYSTEM

1. System Architecture

A distributed AI system can be designed to collect and analyse disaster-related information from multiple sources such as satellite images, IoT sensors and social media.



2. Data Collection and Processing

The system continuously receives data from different sources. Sensor data may contain water level, rainfall, temperature or ground vibrations. Satellite data provides information about affected geographical regions, while social media provides real-time reports from people.

PySpark RDDs divide the large volume of incoming data across multiple machines and process it in parallel. Operations such as `map()`, `filter()`, and `reduceByKey()` can be used for cleaning, transformation and aggregation.

3. Data Fusion

Data fusion combines information from different sources to improve disaster detection.

For example:

High Water Level + Heavy Rainfall + Flood Reports



High Flood Risk

The system can assign confidence scores to different sources and combine them to determine the overall disaster risk.

4. AI-Based Disaster Detection

Machine-learning models analyse the processed data and classify the situation as:

- Normal
- Low Risk
- Medium Risk
- High Risk

For example, abnormal sensor readings combined with satellite evidence can indicate a possible flood even when information from one source is incomplete.

6. Role of Intelligent Agents

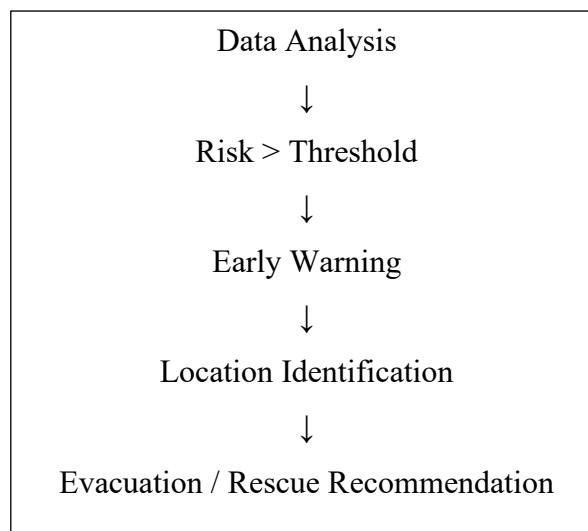
- **Monitoring Agent:** Continuously monitors sensors, satellite information and social media.
- **Analysis Agent:** Analyses the collected data and identifies possible disasters.
- **Warning Agent:** Generates alerts when the risk crosses a specified threshold.
- **Response Agent:** Suggests actions such as evacuation, road closure and deployment of rescue teams.

The agents communicate with each other and coordinate their decisions to provide a faster response.

6. Early Warning and Response

When the AI system detects a high-risk situation, the warning agent can immediately generate an alert.

The system can provide location-based warnings to emergency authorities and affected communities.



7. Advantages

- **Scalability:** PySpark can process large amounts of disaster data using distributed computing.
- **Real-Time Detection:** Continuous data processing allows disasters to be identified quickly.
- **Improved Accuracy:** Combining multiple data sources reduces dependence on a single source.
- **Coordinated Response:** Intelligent agents can divide tasks and coordinate emergency actions.

Conclusion

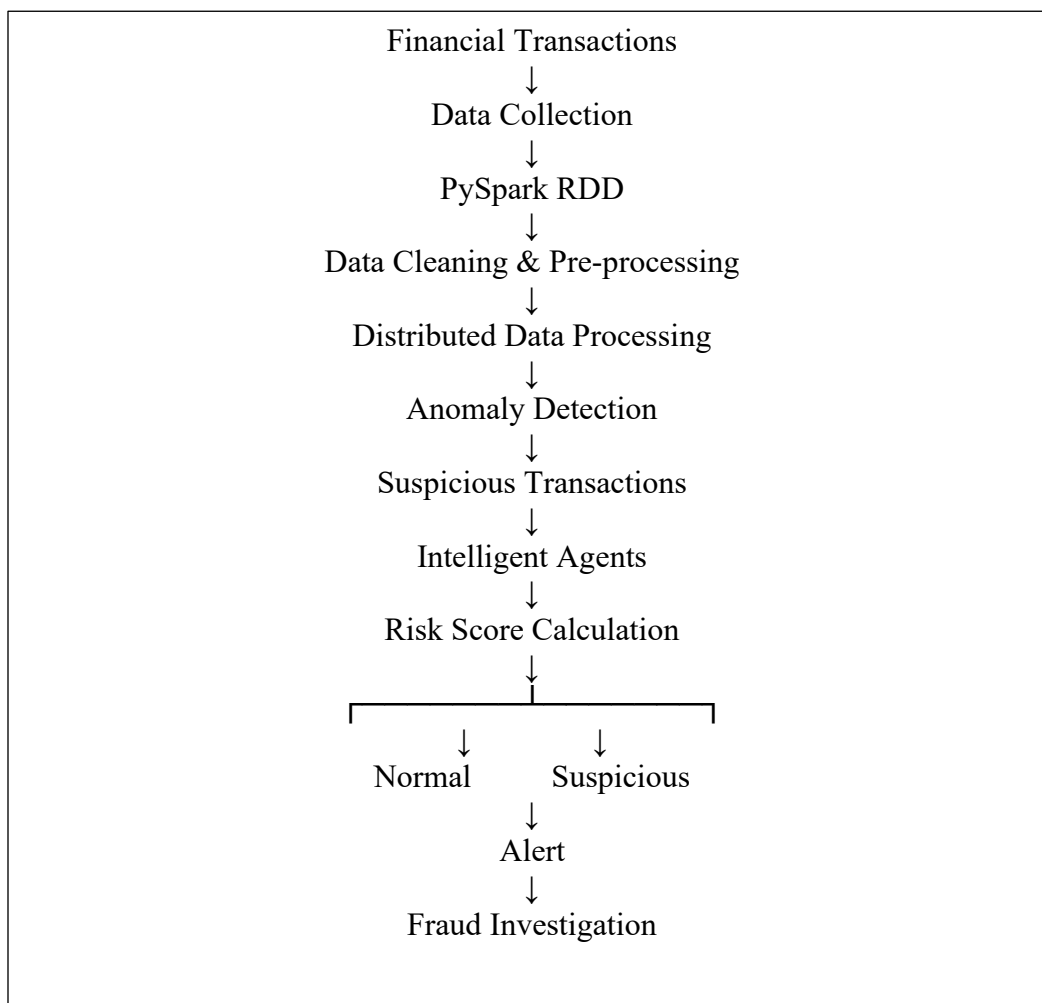
The Smart Disaster Detection System combines PySpark distributed processing, RDD operations, data fusion, AI models and intelligent agents to detect natural disasters quickly. By analysing satellite, sensor and social-media data together, the system can provide early warnings and suitable response strategies, helping authorities reduce damage and improve public safety.

QUESTION 3:-

AI-BASED FRAUD DETECTION FRAMEWORK

1. System Architecture

The proposed system uses PySpark RDDs to process large volumes of financial transactions and identify suspicious activities.



2. Data Processing Using PySpark RDDs

Transaction data may contain details such as transaction amount, location, time, account information and transaction frequency.

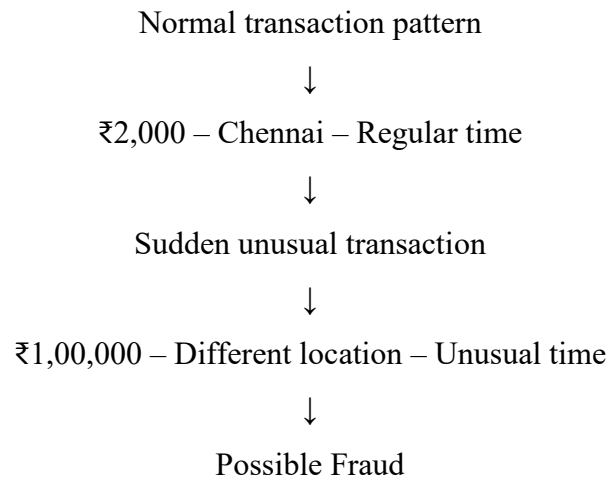
PySpark RDDs distribute these transactions across multiple machines. Operations such as `map()`, `filter()`, `reduceByKey()` and `groupByKey()` can be used to clean data, calculate transaction statistics and identify unusual behaviour.

This distributed approach allows millions of transactions to be processed efficiently.

3. Anomaly Detection

Anomaly detection identifies transactions that significantly differ from normal customer behaviour.

For example:



Algorithms such as **Isolation Forest**, **clustering or statistical anomaly detection** can be applied to identify unusual transaction patterns.

In a distributed environment, the data can be divided into partitions, anomalies can be detected in parallel, and the results can then be combined.

4. Intelligent Agent Collaboration

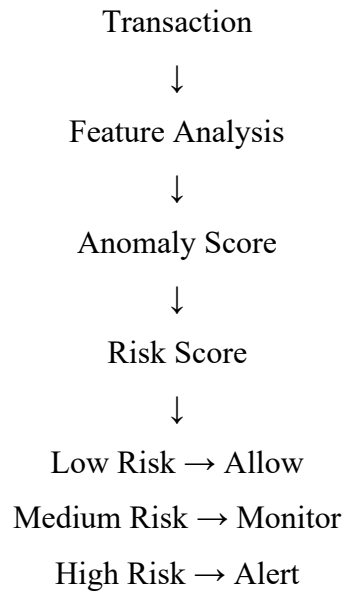
Different intelligent agents can perform different responsibilities:

- **Monitoring Agent:** Continuously monitors incoming transactions.
- **Analysis Agent:** Detects unusual transaction patterns.
- **Risk Agent:** Calculates the fraud risk score.
- **Alert Agent:** Generates alerts for high-risk transactions.
- **Learning Agent:** Uses confirmed fraud cases and feedback to improve the detection model.

The agents communicate their results to make faster and more reliable decisions.

5. Alert Generation

Each transaction can be assigned a risk score based on factors such as transaction amount, location, frequency and unusual behaviour.



High-risk transactions can be flagged for further investigation or additional verification.

6. Continuous Learning

The system continuously receives feedback from fraud investigators and newly confirmed fraud cases. The learning agent uses this information to update the detection model.

This allows the system to adapt to new fraud techniques and reduce false positives over time.

7. Advantages

- **Scalability:** PySpark can process millions of transactions using distributed computing.
- **Fast Detection:** Transactions can be analysed efficiently across multiple nodes.
- **Adaptability:** Intelligent agents can learn from newly detected fraud patterns.
- **Improved Accuracy:** Combining multiple detection methods and agent decisions can improve fraud identification.

Conclusion

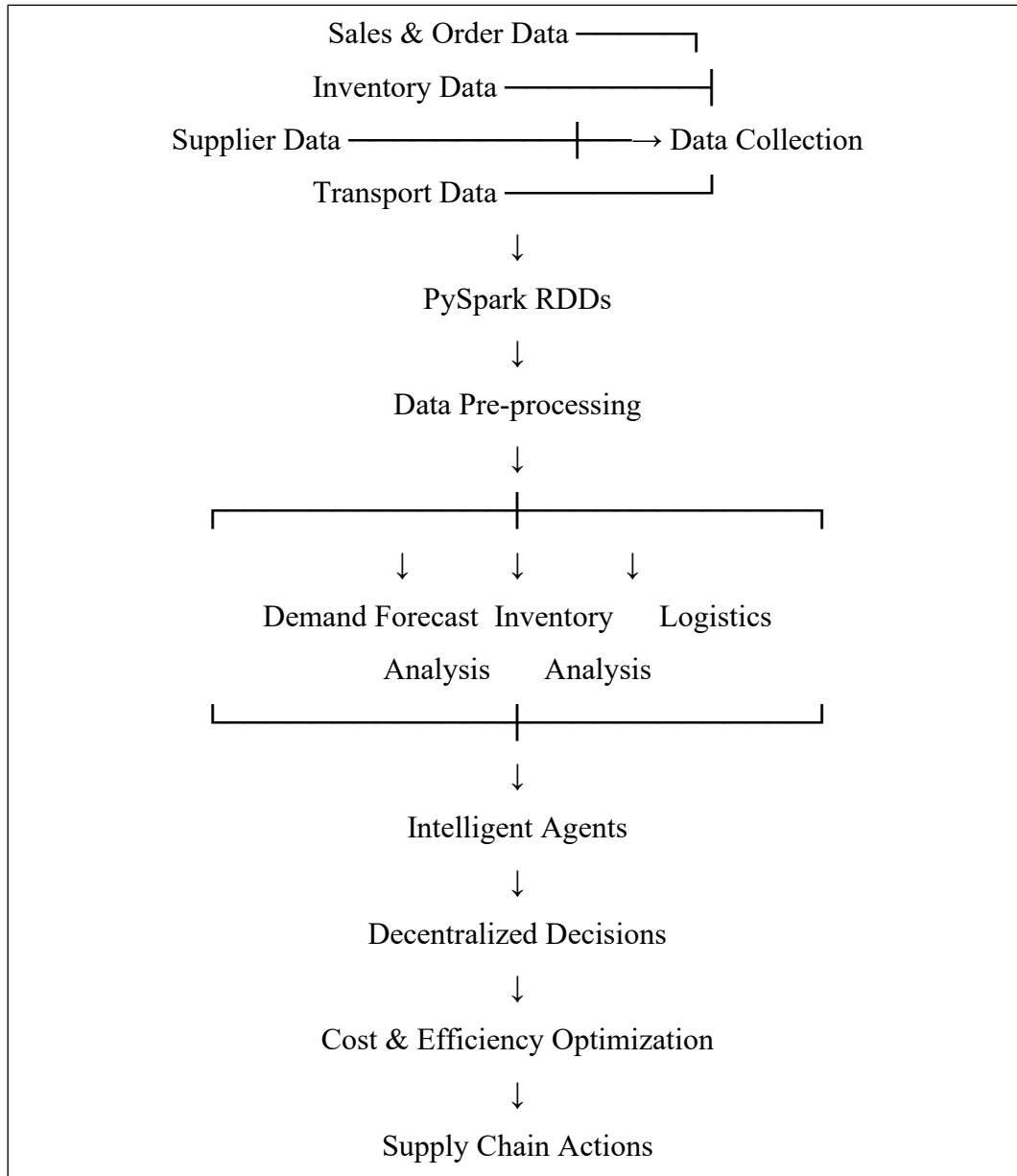
The AI-Based Fraud Detection Framework combines PySpark RDDs, distributed anomaly detection and intelligent agents to identify suspicious financial transactions. PySpark provides scalability for large datasets, while intelligent agents monitor transactions, analyse patterns, generate alerts and learn from feedback. This makes the system suitable for real-time and large-scale financial fraud detection.

QUESTION 4:-

AUTONOMOUS SUPPLY CHAIN OPTIMIZATION SYSTEM

1. System Architecture

The proposed system uses PySpark and intelligent agents to optimize inventory, transportation and demand forecasting across the supply chain.



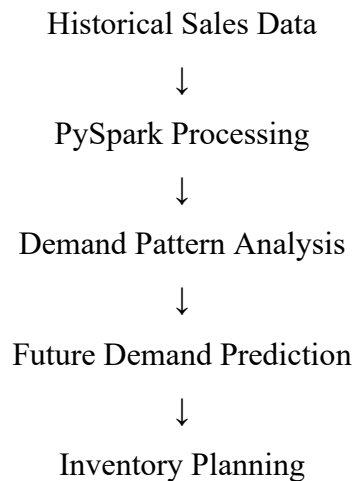
2. Data Processing Using PySpark RDDs

The system collects large amounts of data such as customer orders, inventory levels, supplier information, transportation costs and delivery times. PySpark RDDs divide this data across multiple computers and process it in parallel. Transformations such as `map()`, `filter()`, `flatMap()` and `reduceByKey()` can be used to clean data,

calculate demand and aggregate inventory information. For example, sales records can be grouped by product and location to identify demand patterns.

3. Demand Forecasting

Historical sales data is analysed to predict future demand.



Accurate forecasting helps prevent both overstocking and stock shortages.

4. Intelligent Agent System

- **Inventory Agent:** Monitors stock levels and decides when replenishment is required.
- **Demand Agent:** Analyses sales data and predicts future demand.
- **Logistics Agent:** Selects efficient transportation routes and delivery schedules.
- **Supplier Agent:** Evaluates suppliers based on cost, availability and delivery time.
- **Coordination Agent:** Combines the decisions of different agents and resolves conflicts.

For example, if the demand agent predicts increased demand, the inventory agent can request additional stock while the logistics agent selects a suitable delivery option.

5. Decentralized Decision-Making

Each agent makes decisions according to its local information and objective, while communicating with other agents.



This reduces dependence on a single central decision-maker and allows faster responses to changing conditions.

6. Cost Minimization

The system aims to minimize:

- Inventory holding cost
- Transportation cost
- Storage cost
- Stock-out cost
- Delivery delays

The agents compare different decisions and select actions that provide a good balance between cost and efficiency.

7. Advantages

- **Scalability:** PySpark can process large-scale supply chain data efficiently.
- **Faster Processing:** RDD transformations allow parallel processing across multiple machines.
- **Adaptability:** Agents can respond to changes in demand, inventory and transportation conditions.
- **Cost Reduction:** Coordinated decisions can reduce unnecessary inventory and transportation expenses.
- **Improved Efficiency:** Better forecasting and logistics planning improve delivery performance.

Conclusion

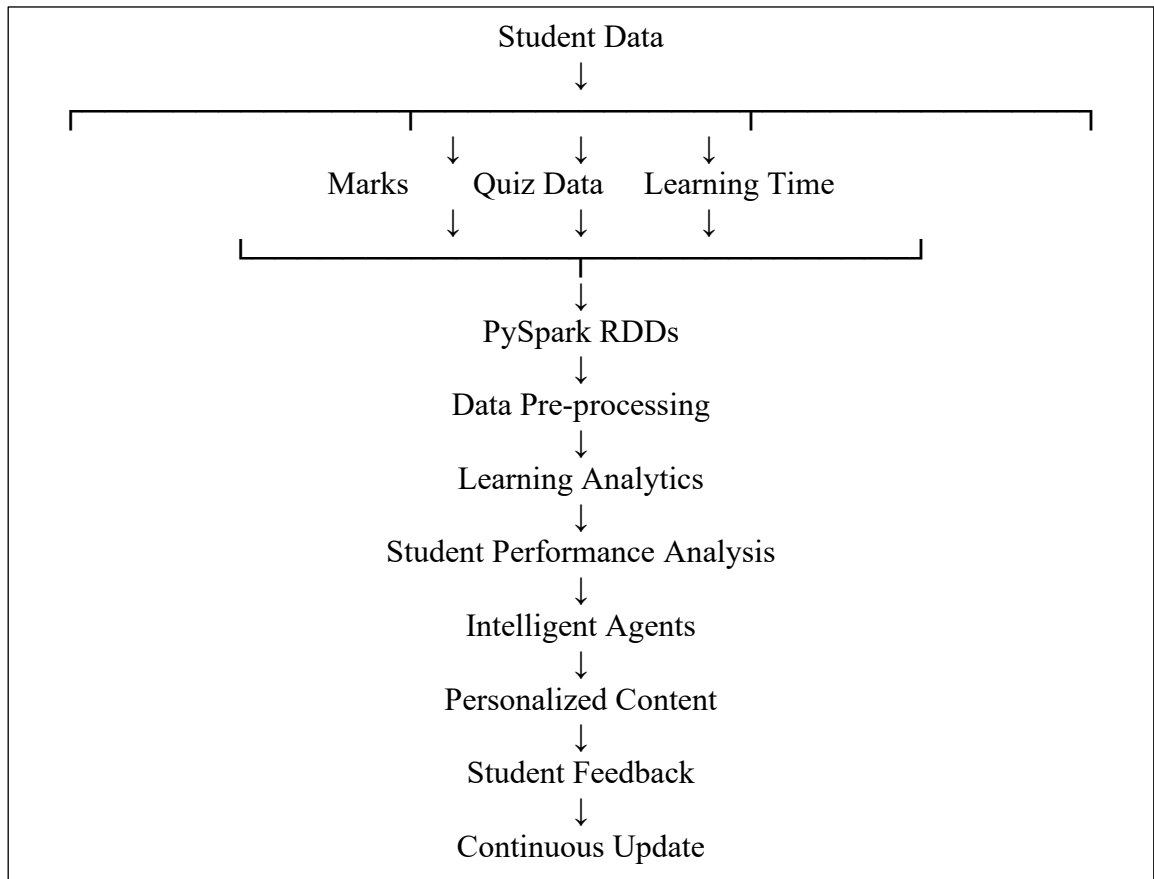
The Autonomous Supply Chain Optimization System combines PySpark distributed processing, demand forecasting and intelligent agents to improve supply chain operations. RDD transformations allow large datasets to be processed efficiently, while multiple agents make decentralized decisions for inventory, suppliers and logistics. By coordinating these decisions, the system can minimize operational costs, reduce delays and improve overall supply chain efficiency.

QUESTION 5:-

PERSONALIZED LEARNING ANALYTICS PLATFORM

1. System Architecture

The proposed system uses PySpark, RDDs and intelligent agents to analyse student learning behaviour and provide personalized learning content.



2. Data Processing Using PySpark RDDs

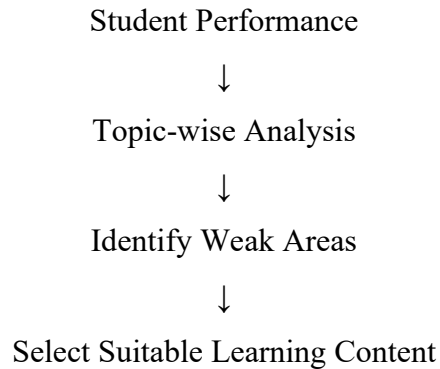
The platform collects large amounts of student data such as quiz scores, assignment marks, attendance, learning time, completed topics and interaction history.

PySpark RDDs distribute this data across multiple computers and process it in parallel. Transformations such as `map()`, `filter()`, `groupByKey()` and `reduceByKey()` can be used to calculate student performance and identify learning patterns.

This makes the system suitable for handling data from thousands of learners.

3. Student Performance Analysis

The system analyses each student's learning behaviour and identifies their strengths and weaknesses.



For example, if a student performs poorly in a particular topic, the system can recommend additional explanations, examples and practice questions.

4. Role of Intelligent Agents

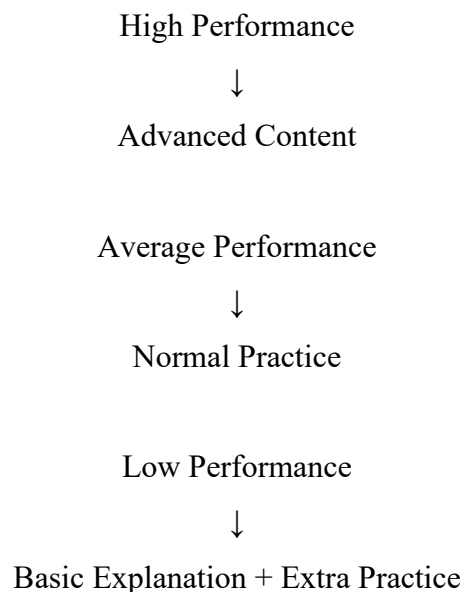
Different intelligent agents can work together:

- **Monitoring Agent:** Tracks student activities and learning progress.
- **Performance Agent:** Analyses marks, quiz results and learning behaviour.
- **Recommendation Agent:** Selects suitable lessons and practice materials.
- **Feedback Agent:** Provides immediate feedback after quizzes and activities.
- **Adaptation Agent:** Continuously updates the learning path based on student progress.

These agents allow the platform to adapt to individual learning needs.

5. Adaptive Learning

The system changes the difficulty and type of content according to the student's performance.



Thus, students do not have to follow exactly the same learning path.

6. Real-Time Feedback

When a student completes a quiz or activity, the system can immediately analyse the result and provide feedback.

For example, repeated mistakes in a topic can trigger a recommendation for revision material. Good performance can lead to more challenging questions.

7. Scalability

PySpark allows the platform to process student records in parallel across multiple machines. Therefore, the system can handle data from thousands of students without depending on a single computer for all processing. This makes it suitable for universities, online learning platforms and large educational institutions.

8. Advantages

- **Personalization:** Learning content is selected according to individual performance.
- **Scalability:** PySpark handles large volumes of student data efficiently.
- **Real-Time Feedback:** Students receive immediate performance-based suggestions.
- **Adaptability:** Intelligent agents continuously update learning recommendations.
- **Better Learning Outcomes:** Identifying weak areas helps students focus on topics that need improvement.

Conclusion

The Personalized Learning Analytics Platform combines PySpark, RDD-based distributed processing and intelligent agents to analyse student learning behaviour at scale. The system provides real-time feedback and adapts content according to individual performance. By processing data from thousands of learners efficiently, it can deliver a personalized, adaptive and scalable learning experience.